

Executive Brief

Two recurring IT problems, two different fixes, and the metrics that will prove them, drawn from 20,769 real ServiceNow incidents.

This analysis answers three questions most IT teams cannot: which problems keep coming back, who should own them, and how we will know a fix worked.

Why this exists

This started with a question from an internship at Protiviti: if incidents are only ever handled verbally, with nothing recorded, how would anyone know if the same problem kept happening? This project is my attempt to answer that, using a separate, real, anonymised, public ServiceNow dataset, not client data. Full story:

[02_End_to_End_Case_Study.pdf](#), section 2.

20,769

incidents analysed
(one row each)

39.4%

overall SLA
breach rate

58 days

Category 34
average resolution

1

record flagged
open, not a real ticket

Figures reflect the corrected final-row model (DQ-08, resolved). See "Data quality" on page 3.

What we found

- **Normal performance hides pockets of failure.** Category 34 takes about 58 days to resolve (439 incidents); Category 46 takes about 14 days. Groups 10, 37 and 31 breach SLA on 74 to 86% of incidents against 39.4% overall, while Group 70 handles 7,369 incidents at only 18.6%. Volume alone does not cause breaches.
- **Recurrence can only be measured by caller and category.** The asset field is empty on 99.75% of incidents, so the original plan (same asset breaking twice) was dropped. Under the caller + category rule there are 73,359 matching pairs within 14 days, and two patterns stand out.
- **The two patterns need different fixes.** One is an ownership problem, the other an automation opportunity. Treating both as "a slow category" would have missed that.

Recommendations

Pattern	What the data shows	Intervention	Proposed owner	Success metric (baseline)
Category 46 1,993 incidents	Spread over 37 groups + unassigned; top group 15%. Reassignments near average (0.92 vs 0.99): not distinctive. Reopens near zero, so fixes hold. Breach 54.1% vs 39.4% overall — this is the real signal.	Routing and ownership: define an owning team and a triage rule.	Service Desk / IT Governance	SLA breach rate (54.1%): should fall toward the overall 39.4%
Caller 1904 / Category 51 399 incidents	All handled by Group 64. Zero reassignments, zero reopens. May be a system or service account; validate.	Automation opportunity: automate resolution or tune the alert.	Automation / Platform team with Group 64	Incident volume (399): should fall sharply toward zero

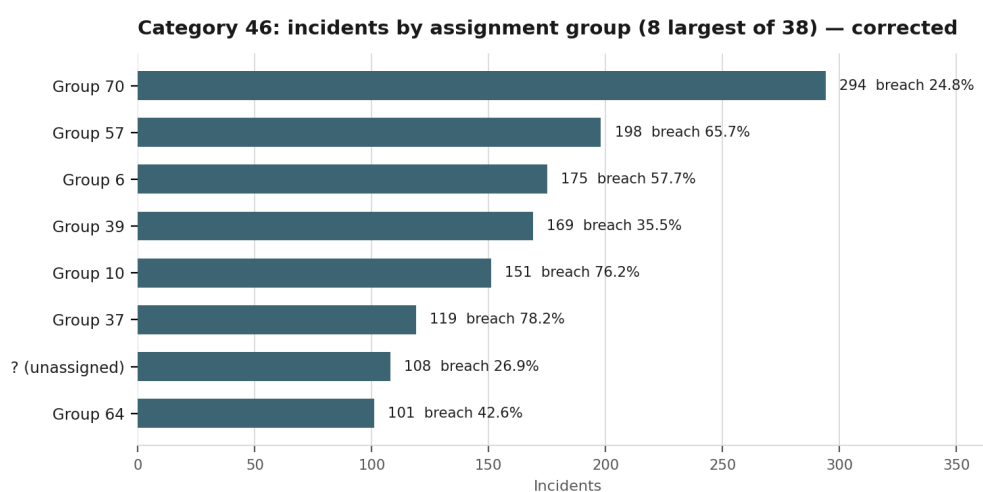
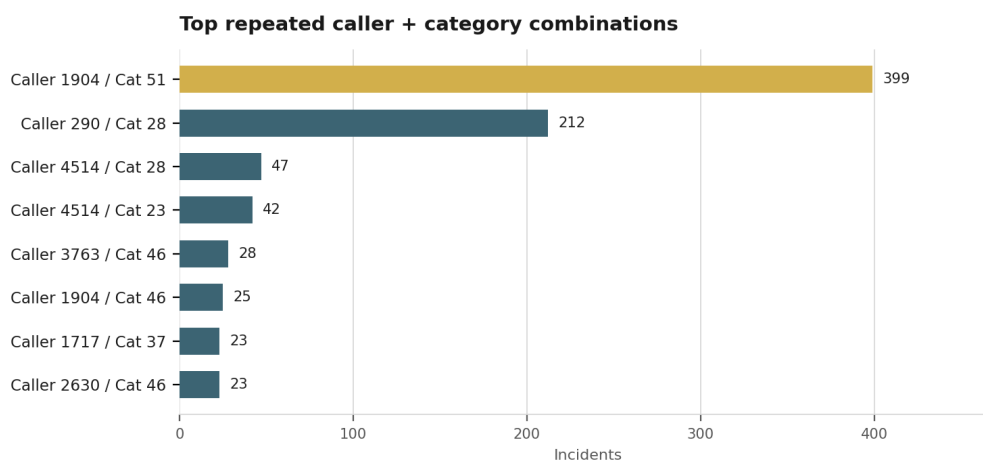
Decisions requested

(1) Nominate an owner and a triage rule for Category 46. (2) Confirm whether Caller 1904 is a person or an automated source. (3) Adopt the two KPIs above and review them after one operating cycle. If they do not move, the pattern is revisited.

How confident are we?

Finding	Evidence	Confidence	What would change our mind
Category 46 has an ownership problem	37 groups + unassigned; top share 14.8% (lowest of any 100+ category); reassignments average; reopens ~0	Medium: association, not proof	Service desk shows a clear owner exists, or the breach rate is explained by something else
Caller 1904 / Category 51 is automatable	399 tickets, one group, zero rework, about two months of sustained volume	Medium to high on the pattern; source unconfirmed	Caller 1904 turns out to be a person with a real recurring fault
Category 34 is a resolution-time outlier	58 days average across 439 incidents, far above every other high-volume category	High: verified on the corrected model	Little to none: barely moved under the DQ-08 fix (~58 days both)

Evidence at a glance



Top: Caller 1904 with Category 51 is almost double the next combination. Below: Category 46 incidents by group, corrected model (24.8% to 78.2% breach rate).

Limits you should know

- Recurrence means the same caller raising the same category, not the same asset failing again.
- Findings are hypotheses. The data shows association, and each pattern lists what the service desk must validate.
- One data-model issue (DQ-08) has been found and resolved: the model was picking each incident's latest row by sorting text dates instead of parsing them. Every backlog, SLA and resolution figure in this document now uses the corrected model; see [02_End_to_End_Case_Study.pdf](#), section 5, for the full before/after comparison.
- The 7-day and 30-day recurrence windows have been run. Recurrence is stable across all three windows and across both final-row models (73,339 to 73,359 matching pairs at 14 days).

Where to go next

[02_End_to_End_Case_Study.pdf](#) for the full story. [03_Analytical_Appendix.pdf](#) for every query result. [dashboard/](#) for the three-page Power BI report. [sql/](#) and [python/](#) to reproduce every number.